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EXTENDED ABSTRACT

Increasing atmospheric vertical resolution in data-driven weather forecasting models

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1. Introduction and objectives

In numerical weather prediction (NWP) models, high vertical resolution is essential for accurately representing atmospheric structures and producing skilful forecasts. In contrast, data-driven weather forecasting models are able to produce skilful forecasts using relatively low vertical resolution.

While this lower resolution is sufficient for many applications, certain users, e.g. in the aviation sector, require forecasts at a much finer vertical resolution. Higher vertical resolution is also required for generating lateral boundary conditions to drive limited-area (LAM) models, as well as for hybrid forecasting systems that use nudging of a physical model [3] or data assimilation, because it enables seamless integration with NWP models.

2. Methods and data

In this work, we test three different approaches to increase the vertical resolution of AIFS, a data-driven global weather forecasting system developed at ECMWF. AIFS is based on an attention-based graph neural network with an encoder–processor–decoder structure [1] implemented in the Anemoi framework (https://anemoi.ecmwf.int/). All methods for increased vertical resolution are first trained on ERA5 reanalysis data and then fine-tuned on operational analysis data from ECWMF's IFS system.

Our first approach is to train AIFS directly on a larger number of vertical levels. In operational mode, AIFS is trained on 13 pressure levels, but here, we train a model, AIFS-ml, on the full set of 137 IFS model levels. The architecture of AIFS-ml is largely the same as that of AIFS-Single v1 [1], but the increased number of input fields (an increase of a factor of 10) means we increase the number of channels in the processor from 1024 to 1536 channels to increase the capacity of the model to learn relationships in the latent space.

Our second approach is a *Vertical-Interpolator* model that takes as input the AIFS-Single v1 pressure level forecasts and outputs the 137 model levels of IFS at the same time step. The model uses an encoder-processor-decoder structure like AIFS, but with only two layers in the processor. As a result, the Vertical-Interpolator requires significantly fewer computational resources to train.

Finally, several studies have shown that the latent space of data-driven weather forecasting models often encodes more structure than is directly expressed in the predicted variables, owing to their ability to learn implicit representations from data [2]. Thus, we develop a third approach, a *multi-decoder*

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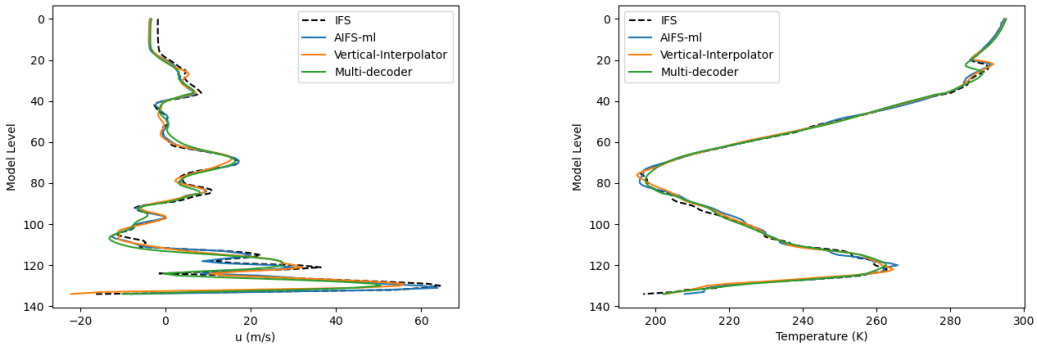

(a) u-component of wind speed at 24h lead time

(b) Temperature at 240h lead time

Figure 1. *Vertical profiles at (4W, 14S) on 1st June 2023 comparing our three approaches with IFS model levels. Note the vertical interpolator is applied to the AIFS-Single v1 forecast.*

where we take a pre-trained AIFS-Single v1 model and fine-tune an extra decoder to target model levels, thus decoding greater resolution from the AIFS latent space. This approach has the advantage of building on a forecast model of known high quality while providing additional information of direct relevance to users.

3. Preliminary results and/or expected outcomes

Each of the three approaches has distinct advantages and limitations. AIFS-ml has been successfully used in a research experiment to spectrally nudge the IFS towards AIFS, producing a hybrid system that combines AIFS's large-scale skill with the physics-based realism of IFS at small scales [3]. However, the increased model size and memory footprint due to the increase in input fields makes this approach computationally very expensive, especially at high horizontal resolutions. Figure 1 compares the vertical profiles from all three approaches at a fixed location in space and time for a complex vertical situation. In general, the Vertical-Interpolator approach is computationally significantly cheaper and preliminary results show generally good performance for the u- and v-components of wind throughout the atmospheric column. However, temperature is not well represented, particularly in the boundary layer and upper stratosphere/lower mesosphere, although the Vertical Interpolator performs well for the latter in our case study. Preliminary results for the multi-decoder approach show similar strengths to the Vertical Interpolator for wind, but for temperature the multi-decoder is better in the boundary layer and worse in the upper stratosphere/lower mesosphere.

All three approaches are promising avenues for increasing vertical resolution, but further work is clearly required, in particular to improve temperature reconstruction.

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